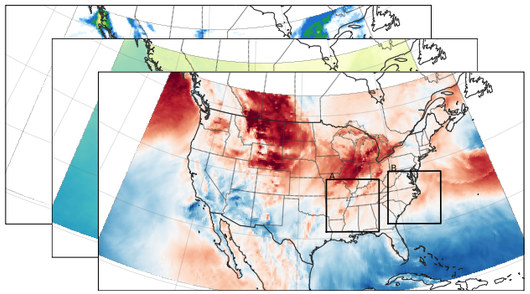
MERRA21C-ML: GENERATIVE CONUS DOWNSCALING

GEOS-FP (~25 km) → Conditional flow matching → GEOS grid (~3 km)

A COARSE INPUTS

GEOS-FP · ~25 km



Weather at t − 1 h and t
Rain rate · PRECTOT
10 m winds · U10M, V10M
Column vapor · TQV
2 m humidity · QV2M
Sea-level pressure · SLP

Coarse baselines at t
T2M, PRECTOT, U10M, V10M
Surface pressure · PS

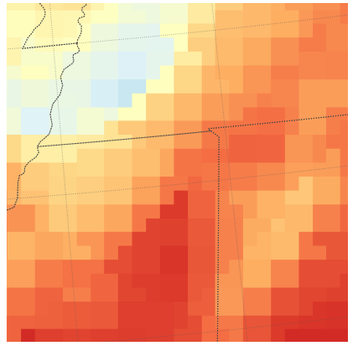
Static fields
Elevation · lat/lon · cell area
Land/lake/ocean fractions
Water distance · x/y slopes
Lake-availability mask

Time (sine/cosine)
Annual · UTC · local solar

B RESOLUTION

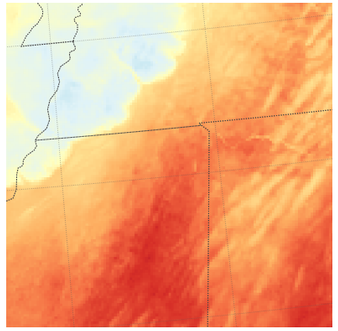
Same extent · same scale

GEOS-FP · ~25 km





GEOS replay · ~3 km



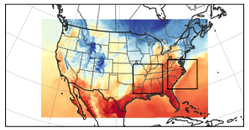


Temperature (°C) · 3× zoom

C GENERATED FIELDS

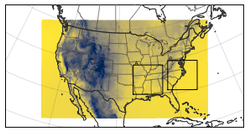
One ensemble member · ~3 km

T₂m (°C)



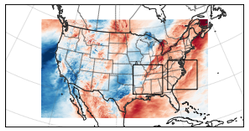


pₛ (hPa)



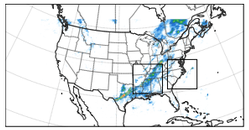


v₁₀ (m s⁻¹)



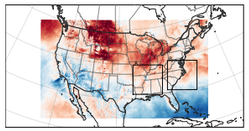


Rain (mm h⁻¹)



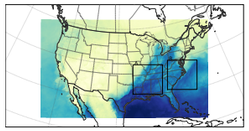


u₁₀ (m s⁻¹)





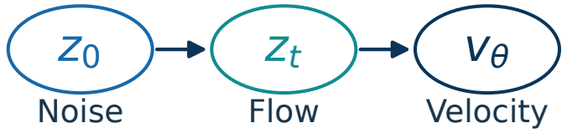
q₂m (g kg⁻¹)



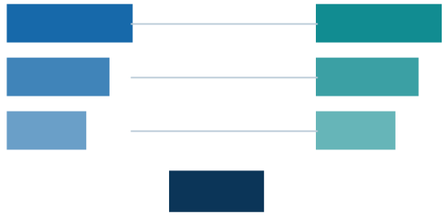


D FLOW MATCHING

Frozen regression conditioner



Conditional U-Net



Weighted velocity loss



Core + halo
Area + channels

Ensemble sampling



Independent noise seeds